Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/06/2005
PI Name:	Patnaik, Gopal Ph.D.		
Project Title:	Unsteady Multidimensional Numerical Simulations of Flame Vortex Interactions in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Washington	**State:**	DC
Zip Code:	20375-5344	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	03/21/2001	**End Date:**	03/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: original end date was 12/31/2004		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schwer, Douglas Alan (Naval Research Lab)		
Grant/Contract No.:	C-32097-N		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A detailed time-dependent flame model is used to isolate and study the underlying controlling physical mechanisms in flame-vortex interactions in reduced gravity and in earth-gravity, i.e. perform complete simulations of current microgravity flame-vortex experiments, investigate the vortex ring generation process to determine size and strength, and study extinction in flame-vortex interactions. Our approach is to use a unique capability developed at NRL over the years to simulate flames in different gravity environments using a detailed time-dependent, multi-dimensional, multi-species numerical model, FLAME3D. The model includes a detailed chemical kinetics mechanism coupled to algorithms for convection, thermal conduction, radiation, viscosity, molecular and thermal diffusion and external forces. The external force, gravity, can have a range of values. The objective is to study the structure (multidimensional) and dynamics (propagation, stability and extinction) of the interaction of vortex rings with premixed flames in micro- and normal-gravity environments Small amounts of fuel leaking from fuel cells could pose a fire hazard on the space-shuttle		

	and similar vehicles. Therefore, for the safe operation of space vehicles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the burning and extinguishment characteristics of flames are important on Earth.
Task Progress:	Work on the flame-vortex interaction project during this year has focused on the development and evaluation of systematically reduced kinetic-mechanisms for accurately simulating flame vortex-interaction in different gravitational environments. The systematically reduced mechanisms were incorporated into the FLAME3D code developed previously. The main idea behind systematic reduction of mechanisms is to reduce a "starting" reaction mechanism by computing some of the intermediate species assuming quasi-steady-state (QSS) behavior. By doing this, we no longer need to solve a transport equation for these intermediate species, and do not include them in the energy balance or the transport property calculations. Systematically reduced mechanisms are detailed in numerous papers (Smooke, 1991) and will only be outlined here. The first step is to choose an appropriate starting mechanism for the given conditions you are interested in. For our studies, we used the methane Peters & Rogg (1993) mechanism as our starting mechanism. Intermediate species are then identified that can be approximated as QSS species. This is typically done by analyzing results using the starting mechanism and comparing temporal and spatial fluxes for candidate species. Once the QSS species are identified, a global mechanism is developed by determining the primary reaction chains and eliminating the fastest reactions using the steady-state approximations. Reaction rate for the global reactions are expressed as a summation of elementary reaction rates. The final step is to develop expressions for the QSS species using both the QSS assumption and also assuming specific reactions are in equilibrium. The systematically reduced mechanism used for our current research follow the mechanisms used in Peters and Rogg (1993) and Chelliah (1993) for methane premixed and diffusion flames. These mechanisms use a four step global mechanism involving the transported species CH4, O2, H2O, CO, CO2, H, H2 and the QSS species OH, O, CH3, CH2O, CH2, HCO, HO2, and H2O2. We adjusted some of the expressions for the QSS species based on a series of one-dimensional premixed flame calculations computed using conditions similar to the flame-vortex conditions. We found that for most conditions we could obtain flame speeds to within 5% of the flame speed calculated using the starting mechanism, although we also found that the reduced mechanisms are generally valid for a much smaller range of conditions than the starting mechanism. After the series of one-dimensional computations, we computed a series of two-dimensional computations with the flame-vortex interaction. In particular, we wanted to reproduce a series of cases that had been accomplished previously with weak, moderate, and strong vortices. The main emphasis of the research was to determine whether we could reproduce similar behavior with the reduced mechanisms as we saw with the starting mechanisms such as flame pinch. From the studies, two main disadvantages were found with using systematically reduced mechanisms. The first disadvantage is that the range of conditions for which a given mechanism is valid is much smaller than with the starting mechanism. Care must be taken to ensure that the mechanism is valid in different regions of the flame. This can be difficult, especially for complex multi-dimensional flow-fields. Secondly, the QSS assumptions break down outside of the actual flame region. Because of this, the reduced mechanisms are less robust than the starting mechanism. In particular, careful attention must be given to generating valid initial conditions whether that initial condition is important or not in the subsequent studies. In all cases, we were able to reproduce similar phenomena with the reduced mechanisms as was seen with the original starting mechanism in previous studies. The amount of CPU savings was about a factor of two better than the starting mechanism. However, for larger hydrocarbons, where the number of QSS intermediates should be much larger, there is a potential for much more significant CPU savings.
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